Comment 3: Information retention and alternative subgraph selection

Revision experiment insertion draft

6 November 2026

This standalone preview collects the additional experimental methods, results, and figures. It has not been integrated into the original manuscript or supplementary material. Table/section numbering is provisional.

1 Additional experiments

1.1 Information retention and alternative subgraph selectors

We assess evidence compression at three separate levels: retained graph structure, the representations supplied by the frozen graph encoder, and generated answers. All comparisons fix the candidate identities and their ordering to the original fuzzy input. For repoDB this is the original raw/display-control ordering. Candidate ranking metrics are therefore invariant by construction and are not used as evidence of information retention.

Source graphs and matched budgets. We distinguish the multiset of retrieved edge occurrences from the set of unique directed, relation-labeled triples. Two uncompressed references retain either the full multiset or only unique triples. The original fuzzy graph is read directly from its archived artifact. For each query, all compressed alternatives select exactly the number of unique triples in that artifact from the same unique source-edge set. The empty-graph diagnostic removes local message passing while retaining frozen global entity embeddings. These uncompressed and empty references are not treated as matched-budget competitors.

Alternative selection strategies. The alternatives include uniform random sampling without replacement; random ordering followed by candidate/query coverage; top-B native-score selection without coverage; native-score selection with a strict coverage-then-fill pass; endpoint-degree top-B; query-centered breadth-first ordering; query-personalized PageRank; terminal-personalized PageRank; shortest-path preservation; and relation-stratified selection. Native scores are recovered from archived features/scores, with exact triple and candidate alignment checks. The strict native-score coverage control is not claimed to reproduce every dataset-specific reservation pass of the original implementation.

PageRank uses an unweighted undirected source graph, damping 0.85, and either the query alone or the query and present candidates as uniformly weighted teleportation nodes. Dangling mass returns to the teleportation distribution. Iteration stops at an L1 change below 10^{-10} or after 200 iterations. Edges are ranked by the sum of endpoint scores. The shortest-path selector considers candidates in their fixed order, adds one shortest path of length at most three if the complete path fits the remaining budget, and fills remaining slots by native score. Relation-stratified selection cycles over relation IDs, ranking edges within each stratum by native score. Stable source-index tie-breaking is used unless the explicitly seeded random ordering applies. No selector accesses the held-out gold label.

Five random seeds are evaluated at the native budget. Additional strict-budget curves use 20%, 35%, 55%, 75%, and 90% of unique source edges, with three seeds for random selection. The main CPU study contains 62 variants per dataset/split across all five datasets and both validation and test. A separate 60-row native-implementation audit calls the original selector functions, reproduces the archived edge sets at their native budgets, and varies only the requested budget. It preserves and reports any budget excess caused by the original reservation passes rather than silently imposing a new policy.

Structural retention definitions. Candidate coverage measures incidence to at least one retained edge and is reported both absolutely and relative to source-covered candidates, including top-, middle-, and tail-rank bands. Query coverage, unique-node retention, relation-type retention, and direct query–candidate exposure are reported separately. For $k \in \{2, 3\}$, let $C_{\leq k}(G_q)$ be candidates reachable from the query within k edges in the source graph and let $P_k(G_q)$ be its edge-labeled simple path witnesses of exactly k edges ending at a candidate. We compute

$$R_{\text{reach},k}(q) = \frac{|C_{\leq k}(S_q)|}{|C_{\leq k}(G_q)|}, \qquad R_{\text{path},k}(q) = \frac{|\{p \in P_k(G_q) : E(p) \subseteq E(S_q)\}|}{|P_k(G_q)|}.$$

The walks are traversable in either direction; exact edge identities, relation labels, and traversal signs are retained. Consequently these quantities measure structural connectivity, not necessarily directed biochemical pathways. Typed pattern retention measures whether each source relation/sign sequence of length two or three still has at least one witness. A separate audit measures retention of unique incoming edges within two predecessor steps of the query and candidates, matching the two message-passing layers of the actual directed encoder more closely.

Relation-frequency and node-incidence mass distributions are compared using base-2 Jensen–Shannon divergence. Rare-relation edge recall uses relations with source frequency at or below the per-query median. Explicit Gene annotations from DRKG/repoDB namespaces and unambiguous Hetionet metadata support an additional gene-intermediate path diagnostic; types are not inferred from gene-like names in PrimeKG or PharmKG. Gold-incident and gold-reachability statistics are computed only after selection and only when the gold is in the fixed candidate set and the relevant source denominator is nonzero.

Any retention with a zero source denominator is undefined, omitted from the mean, and accompanied by its eligible sample count. The reported means are averages of per-query-row ratios, not ratios of pooled path counts. Quantiles, complete-loss fractions, source-query-presence strata, and gold-coverage strata supplement the aggregate results. Twenty illustrative cases are selected post hoc by fixed low/median/high structural-retention strata; they are not chosen for favorable generated answers or asserted biological mechanisms.

Frozen-model evaluation and numerical audit. The GPU study evaluates all 22 native-budget variants and six additional strict-budget variants (native-score coverage, terminal PageRank, and random seed zero at 35% and 75%) on the complete test split. The same frozen checkpoint, prompt, candidate order, and deterministic decoding settings apply to every selector within each dataset. We measure cosine similarity and relative L2 change of query/candidate adapter outputs against both source references. Generated-answer exact accuracy, invalid-output rate, top-candidate copying, normalized output agreement, and the explicitly standardized promote-to-front adjusted MRR are reported separately.

A numerical audit found nine nonfinite FP16 encoder outputs for PrimeKG source graphs. FP32 encoder outputs were finite in all 4,200 source/fuzzy test-graph cases audited, but some projected values exceeded the FP16 exponent range. The final protocol therefore loads graph-encoder checkpoint weights in FP32 and LLM weights in BF16, checks finiteness before and after projection casting, and performs neither clipping nor sample exclusion. Earlier FP16 prediction caches are not reused. This precision change is applied uniformly across all methods and is disclosed when comparing to historical results.

The existing encoder falls back to global/global embeddings for graphs with at most ten edge occurrences; we explicitly report this condition at small budgets. An encoder benchmark uses up to 32 distinct query groups, three repetitions, warm-up calls, and CUDA synchronization. It includes graph construction, encoding, projection, and output transfer, but excludes LLM generation and must not be interpreted as end-to-end latency. CPU selection-call timings start after source-context construction and score loading; shared adjacency/path enumeration is excluded and ordering caches can be reused. These descriptive timings are not a total-runtime comparison of selection algorithms.

Uncertainty and scope. Paired query-cluster bootstrap intervals use 2,000 resamples so that rows sharing the same query remain together. Random-selector means additionally receive joint query-cluster/seed bootstrap intervals; seed standard deviations are reported separately. These are exploratory intervals without multiplicity adjustment. All methods, ratios, and seeds are declared before inference; no test-winning configuration is promoted to a new default. Retention is measured relative to retrieved source subgraphs and does not establish preservation of all KG facts, expert-validated mechanisms, or clinical utility.

1.2 Structural retention results and compression trade-offs

The primary CPU study completed 620 dataset/split/variant summaries, with a further 60 summaries from the native-implementation budget audit. All native edge sets were reproduced exactly at their archived budgets on validation and test. Alternative selectors preserved candidate order and satisfied the matched unique-edge budgets. These summaries reuse 2,100 validation and 2,100 test rows; they are not independent new datasets or examples.

Duplicate removal is distinct from unique-edge removal. PrimeKG source graphs contain a mean of 59.932 edge occurrences but only 38.094 unique triples per test row. The original fuzzy output retains 32.340 unique triples. Thus, the mean reduction of 27.592 occurrences consists of 21.838 duplicate occurrences and 5.754 unique triples. Approximately 79.1% of the aggregate count reduction is attributable to duplicate removal. The other four datasets have no such source duplicates in these prepared test inputs. This is a counting decomposition, not a claim that duplicates have no computational effect: the encoder can process repeated triples as repeated messages.

Table 1: Matched-budget test retention of candidate–query reachability within three edges. Coverage is relative to source-covered candidates. Random reports mean ± standard deviation over all five seeds, not a confidence interval. N counts rows with at least one reachable source candidate.

Dataset	N	Fuzzy coverage	Fuzzy reachability	Random	Shortest paths
PrimeKG	488	1.0000	0.7737	0.7760 ± 0.0038	0.9985
PharmKG	500	1.0000	0.9010	0.4423 ± 0.0064	1.0000
Hetionet	90	1.0000	0.4835	0.4848 ± 0.0428	1.0000
DRKG	386	1.0000	0.8921	0.5043 ± 0.0170	1.0000
repoDB	297	1.0000	0.9546	0.5465 ± 0.0241	1.0000

Table 2: Retention of exact edge-labeled simple paths of length three at matched budgets. N is the number of rows containing at least one such source path; rows without source paths are undefined and excluded. These paths are structural witnesses, not validated therapeutic mechanisms.

Dataset	N	Original fuzzy	Random (five seeds)	Shortest paths
PrimeKG	488	0.6543	0.6443 ± 0.0047	0.8199
PharmKG	31	0.0000	0.1801 ± 0.0967	0.8604
Hetionet	64	0.0000	0.1615 ± 0.0589	0.6207
DRKG	201	0.4767	0.1629 ± 0.0078	0.6496
repoDB	219	0.5286	0.1552 ± 0.0185	0.8322

Coverage does not imply path preservation. Although original fuzzy retrieval preserves relative candidate coverage on all five datasets, its reachability retention varies substantially (Table 1). Fuzzy retrieval preserves substantially more reachability than uniform random selection on PharmKG, DRKG, and repoDB, whereas the PrimeKG and Hetionet point estimates are similar to random. The joint query-cluster/seed bootstrap interval for random minus fuzzy reachability is $[-0.4829, -0.4337]$ on PharmKG. The corresponding intervals are $[-0.4433, -0.3353]$ on DRKG and $[-0.4492, -0.3661]$ on repoDB; the PrimeKG and Hetionet intervals include zero. These are exploratory, unadjusted intervals.

The shortest-path baseline retains nearly all source-reachable candidates at the same edge count. This is partly aligned with its explicit selection objective and does not establish semantic superiority. It also trades away coverage of other candidates: relative candidate coverage is 0.8514 on Hetionet, 0.8199 on DRKG, and 0.8971 on repoDB, compared with 1.0000 for original fuzzy retrieval. Thus, preserving candidate incidence and preserving connected query–candidate paths are different selection goals.

Length-three path retention exposes additional losses that are hidden by coverage alone (Table 2). Original fuzzy retrieval retains no such witnesses in the eligible PharmKG and Hetionet rows. This does not mean that every row loses all connectivity: only 31 PharmKG rows have a length-three source witness, and shorter paths can remain. Conversely, original fuzzy retrieval retains more length-three witnesses than random on DRKG and repoDB. The full relation-distribution, rank-band, incoming-context, gene-annotation, and gold-conditional diagnostics are supplied in the result workbook rather than collapsed into a single information-retention score.

Source availability and annotated paths. Fuzzy outputs contain the query node in 100%, 96.8%, 100%, 84.8%, and 59.4% of test rows on PrimeKG, PharmKG, Hetionet, DRKG, and repoDB, respectively. Conditional reachability retention must be interpreted with its source eligibility rather than as a fraction of every test row. The additional Gene-intermediate diagnostic pools length-two and length-three witnesses with explicitly annotated intermediate nodes: fuzzy retention is 1.0000 on the 31 eligible Hetionet rows and 0.9366 on the 243 eligible DRKG rows. These small, annotation-dependent subsets do not establish general mechanism preservation. In particular, the Hetionet value does not contradict the loss of its separate length-three witness set. PrimeKG and PharmKG types are unavailable to this audit; repoDB has no eligible source Gene-intermediate witnesses.

Budget feasibility and operating curves. The native replay audit confirms that the original PrimeKG coverage-first implementation can exceed a newly imposed small target. At a requested budget of 20% of unique source edges, its mean requested/actual counts are 7.576/15.662, and 99.8% of test rows exceed the request. At 35%, the counts are 13.336/15.886 and the excess rate is 74.2%. No excess occurs at the archived native budgets used in the matched comparison. We report these native-policy points separately from strict-budget controls. Operating curves must therefore identify the selector and distinguish requested budget from actual graph size.

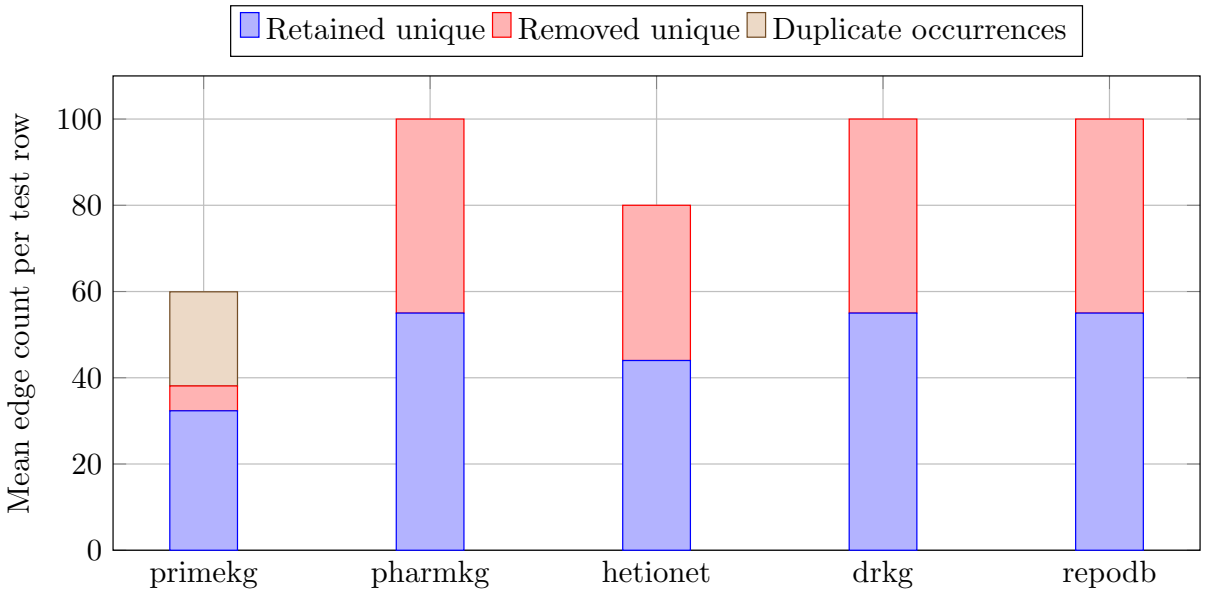


Figure 1: Decomposition of source edge occurrences into retained unique triples, removed unique triples, and duplicate occurrences. Counts are means over test rows; duplicate removal is separated from structural edge loss.

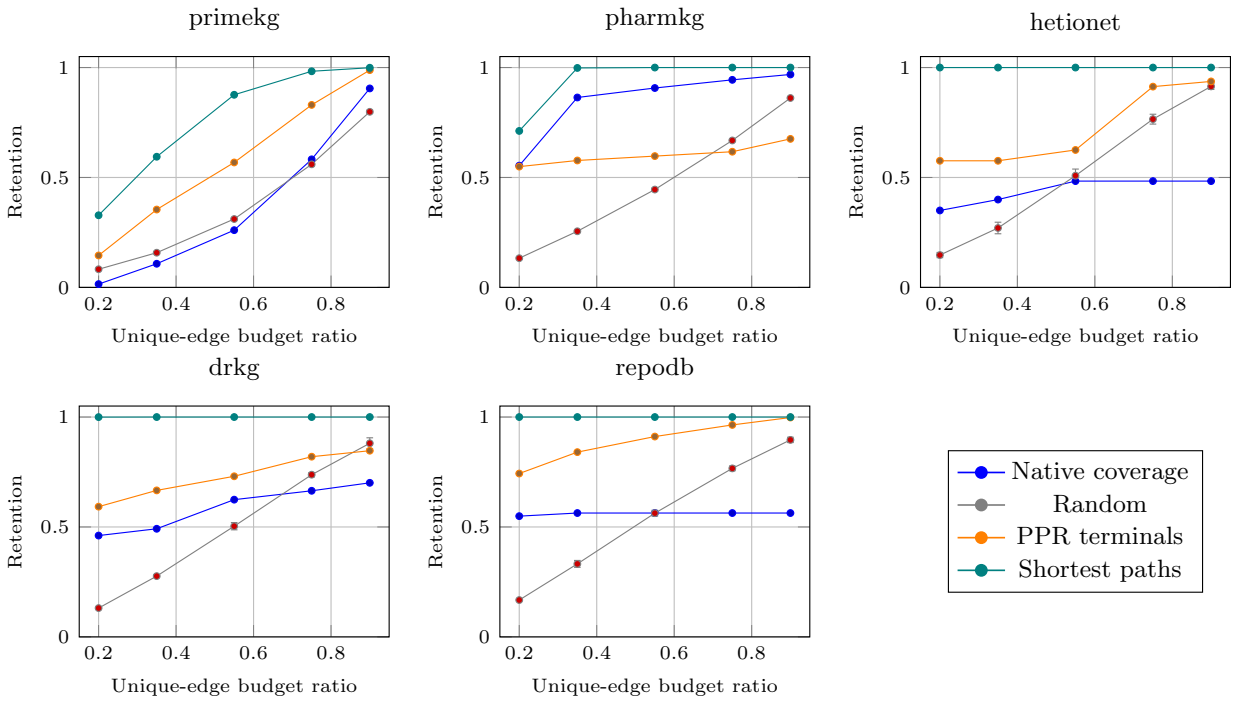


Figure 2: Candidate–query reachability retention for strict-budget selectors. The native-score coverage control is distinguished from the original dataset-specific selector. Random error bars show seed variability, not test confidence intervals. Native implementation budget violations are reported separately in the audit tables.

Table 3: Exact generated-answer accuracy (%) under the standardized matching rule. Random rows average all five seeds. Source raw/unique and empty local graph are diagnostic references with different budgets; the other rows use matched unique-edge budgets. These are new frozen-model diagnostics, not replacements for historical tables computed with other precision or scoring rules.

Method	primekg	pharmkg	hetionet	drkg	repodb
Source raw	0.00	0.00	3.00	1.40	0.20
Source unique	0.00	0.00	3.00	1.40	0.20
Original fuzzy	0.00	0.20	3.00	1.40	0.40
Empty local graph	0.40	0.00	1.00	1.20	1.80
Native score top-k	0.00	0.00	2.00	1.20	2.20
Native coverage	0.00	0.20	3.00	1.40	0.40
Degree top-k	0.00	0.20	0.00	1.20	1.60
Query BFS	0.00	0.20	0.00	1.20	0.40
PPR query	0.00	0.20	0.00	1.20	0.40
PPR terminals	0.00	0.20	2.00	1.20	0.40
Shortest paths	0.00	0.20	3.00	1.40	0.20
Relation stratified	0.00	0.00	2.00	1.20	0.20
Random (5 seeds)	0.04	0.04	3.00	1.24	0.52
Random + coverage (5 seeds)	0.00	0.08	3.00	1.40	0.16

Table 4: Original fuzzy versus the uncompressed source: encoder output changes and normalized generated-answer agreement. Candidate metrics average the 20 candidate embeddings within each row, then average rows. A cosine near one does not exclude a change in vector magnitude; relative L2 is reported separately.

Dataset	Query cosine	Candidate cosine	Query rel. L2	Candidate rel. L2	Answer agreement (%)
primekg	0.8828	0.9453	0.8818	0.1013	78.8000
pharmkg	0.8718	0.9991	0.6130	0.0032	80.0000
hetionet	0.9209	1.0000	0.3259	0.0000	100.0000
drkg	0.9312	0.9867	0.3834	0.0526	100.0000
repodb	0.9814	0.9806	0.1326	0.0862	80.8000

1.3 Frozen-model representation and generated-answer results

The completed GPU study comprises 28 variants on each of five datasets, yielding 140 summaries and 58,800 row–variant evaluations of the same 2,100 test rows. Every comparison uses the same dataset-specific checkpoint and fixed candidate order. The precision policy is FP32 for graph encoding and BF16 for the LLM throughout; no old FP16 predictions are substituted.

Structural loss and output invariance are distinct. On Hetionet, original fuzzy retrieval preserves the candidate encoder outputs (mean cosine 1.0000, relative L2 0.0000) and all generated answers relative to the source graph. Nevertheless, query outputs change (mean cosine 0.9209, relative L2 0.3259), and all eligible length-three path witnesses are lost. Thus even exact candidate-output agreement cannot be interpreted as whole-graph information preservation. Directed message passing, global embeddings, and which nodes are injected into the prompt all affect how a structural change propagates to generation.

DRKG likewise has 100% generated-answer agreement for fuzzy versus source, although mean query and candidate cosines are 0.9312 and 0.9867; their relative L2 changes are 0.3834 and 0.0526. On PrimeKG and PharmKG, answer agreement is only 78.8% and 80.0%, respectively. Their candidate cosines are 0.9453 and 0.9991, while query cosines are lower at 0.8828 and 0.8718. These results show why candidate similarity alone is insufficient to characterize information supplied to the LLM.

On repoDB, fuzzy/source answer agreement is 80.8%, with query/candidate cosines of 0.9814/0.9806. Fuzzy generates 2/500 exact answers, compared with 1/500 for source raw, 11/500 for native-score top-k, and 9/500 for the empty-local-graph reference. The latter retains frozen global embeddings; it is not a knowledge-free model. These counterexamples are reported without selecting a new method using test outcomes.

Downstream differences do not establish a universal winner. The exact-accuracy comparison does not show consistent superiority of fuzzy over all alternative selectors. Several differences correspond to only one correct answer in a 500-row test set. For example, fuzzy produces 0/500, 1/500, 3/100, and 7/500 correct answers on PrimeKG, PharmKG, Hetionet, and DRKG, respectively. The joint query/seed and paired query bootstrap tables are supplied alongside these point estimates; lack of a detected difference is not a demonstration of equivalence. These low counts limit conclusions about small downstream improvements.

Adjusted MRR is intentionally kept separate from exact accuracy. It assigns credit based on the pre-existing candidate order after a predicted candidate is promoted to the front, with zero credit when the gold is absent from the candidate set. It can remain positive when no generated answer is correct. Under the strict normalized matching rule, 59.0% of PharmKG fuzzy outputs do not match a listed candidate. This is an output-format/matching diagnostic, not evidence that those outputs are valid biological alternatives. The full workbook includes invalid-output and top-candidate-copy rates for every variant.

Numerical and computational scope. The FP16 audit failure is localized to nine PrimeKG source-graph cases among the 4,200 source/fuzzy cases checked; the paired FP32 audit contains no nonfinite outputs. The maximum finite FP32 projection magnitude is 130,444.35, exceeding the FP16 maximum of 65,504, motivating the BF16 LLM input policy as well as FP32 graph computation. Final inference checks finiteness before and after casting. All test rows remain included.

The accompanying benchmark reports graph-encoder call timing and resident plus peak allocated CUDA memory, with the LLM already resident but generation excluded. It does not establish an end-to-end LLM speedup. The full predictions, input hashes, model fingerprints, seed results, and completion markers are retained for audit and reproducibility.

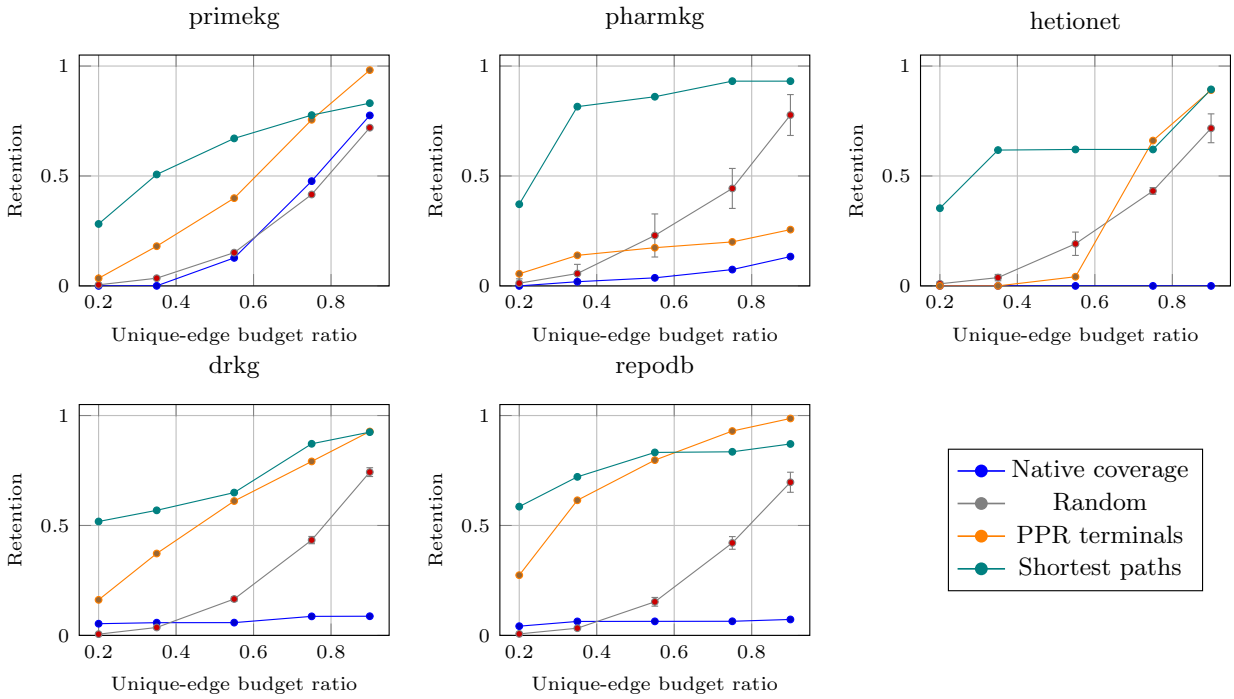


Figure 3: Exact length-three path-witness retention under the same strict-budget controls. Means exclude queries without a corresponding source witness; eligible counts are provided in the accompanying tables.